

Grade-9

Geography

Unit- The Dynamic Atmosphere and the Changing Climate

Reference Material

Atmosphere

Q1. What is the atmosphere?

Answer: The atmosphere is a thick blanket of gases surrounding the Earth. It protects life, regulates temperature, and makes life possible.

Q2. Why is the atmosphere important?

Answer:

- Provides air for breathing.
 - Protects from harmful solar radiation.
 - Regulates Earth's temperature.
 - Makes weather and climate possible.
-

2. Layers of the Atmosphere

Q3. Name the five layers of the atmosphere.

Answer:

1. Troposphere
2. Stratosphere
3. Mesosphere
4. Thermosphere (Ionosphere)
5. Exosphere

Q4. Which layer contains all weather phenomena?

Answer: Troposphere. It contains clouds, rainfall, storms, and lightning.

Q5. What is the lapse rate?

Answer: The temperature decreases by about **1°C for every 165 metres** increase in altitude in the troposphere.

Q6. Why is the stratosphere important?

Answer: It contains the ozone layer which absorbs harmful ultraviolet (UV) rays from the Sun.

Q7. Which layer burns meteorites?

Answer: Mesosphere.

Q8. Why is the thermosphere important?

Answer: It helps in radio transmission, satellite communication and acts as Earth's radiation shield.

Q9. Which is the outermost layer of the atmosphere?

Answer: Exosphere.

3. Composition of Atmosphere

Q10. What are the major gases present in the atmosphere?

Gas	Percentage
Nitrogen	78%
Oxygen	21%
Argon	0.93%
Carbon Dioxide	0.04%

Q11. Which gas is most abundant in the atmosphere?

Answer: Nitrogen (78%).

Q12. Why is oxygen important?

Answer:

- Essential for respiration.

- Supports combustion.
- Necessary for life on Earth.

Q13. What is the importance of carbon dioxide?

Answer:

- Used by plants for photosynthesis.
- Helps maintain Earth's temperature through the greenhouse effect.

Q14. What is the importance of water vapour?

Answer:

- Forms clouds and rainfall.
- Controls humidity.
- Moderates temperature.

Q15. What is the role of ozone?

Answer: Ozone absorbs harmful UV rays and protects living organisms from skin cancer and genetic damage.

4. Weather and Climate

Q16. Differentiate between weather and climate.

Weather	Climate
Day-to-day condition of atmosphere	Average weather over a long period
Changes frequently	Changes slowly

Q17. What are the main elements of weather and climate?

Answer:

1. Temperature
2. Atmospheric Pressure
3. Wind
4. Humidity
5. Precipitation

5. Temperature

Q18. What is temperature?

Answer: Temperature is the degree of hotness or coldness of air.

Q19. Which instrument measures temperature?

Answer: Thermometer.

Q20. Formula for Temperature Range

Answer:

Temperature Range = Maximum Temperature – Minimum Temperature

Q21. Formula for Mean Daily Temperature

Answer:

(Maximum Temperature + Minimum Temperature) ÷ 2

6. Atmospheric Pressure

Q22. What is atmospheric pressure?

Answer: It is the force exerted by the weight of air pressing on the Earth's surface.

Q23. Which instrument measures atmospheric pressure?

Answer: Barometer.

Q24. What is the average atmospheric pressure at sea level?

Answer: 1013 millibars (mb) or 1 ATM.

7. Wind

Q25. What is wind?

Answer: Wind is the movement of air from high-pressure areas to low-pressure areas.

Q26. Which instrument measures wind speed?

Answer: Anemometer.

Q27. Which instrument shows wind direction?

Answer: Wind vane.

8. Humidity

Q28. What is humidity?

Answer: Humidity is the amount of water vapour present in the air.

Q29. What is relative humidity?

Answer: It is the percentage of moisture present in air compared to the maximum moisture it can hold at that temperature.

Q30. Which instrument measures humidity?

Answer: Hygrometer.

9. Precipitation

Q31. What is precipitation?

Answer: Water falling from the atmosphere to the Earth in the form of rain, snow, hail, etc.

Q32. Which instrument measures rainfall?

Answer: Rain Gauge.

10. Seasons in India

Q33. Name the four seasons of India.

Answer:

1. Hot Weather Season
2. Cold Weather Season
3. Advancing Monsoon Season
4. Retreating Monsoon Season

Q34. When does the Hot Weather Season occur?

Answer: March to May.

Q35. What are Loo winds?

Answer: Strong, hot and dry winds that blow over northern and northwestern India during summer.

Q36. What are Mango Showers?

Answer: Pre-monsoon showers in Kerala and Karnataka that help ripen mangoes.

Q37. Which are the coldest months in India?

Answer: December and January.

Q38. What are Western Disturbances?

Answer: Low-pressure systems originating from the Mediterranean region that bring winter rainfall to northern India.

Q39. What are the two branches of the Southwest Monsoon?

Answer:

1. Arabian Sea Branch
2. Bay of Bengal Branch

Q40. What is October Heat?

Answer: Hot and humid weather experienced during the retreating monsoon season in October.

11. Climate Change

Q41. What is climate change?

Answer: Long-term changes in Earth's temperature and weather patterns mainly caused by human activities.

Q42. What are the major causes of climate change?

Answer:

- Burning fossil fuels
- Deforestation
- Industrial activities
- Agriculture
- Waste generation

Q43. Mention four effects of climate change.

Answer:

- Rising temperatures
 - Melting glaciers
 - Sea-level rise
 - More floods, droughts and storms
-

12. Floods

Q44. What is a flood?

Answer: Overflow of water onto normally dry land due to heavy rainfall, river overflow, cyclones, dam failure, etc.

Q45. What is a flood plain?

Answer: The flat area alongside a river that gets flooded during floods.

Q46. What is a catchment area?

Answer: The upper area that receives rainfall and feeds a river.

Q47. Mention any three effects of floods.

Answer:

- Loss of life and property
- Spread of diseases
- Soil erosion and crop damage

Q48. Mention three flood control measures.

Answer:

- Flood forecasting and warning
 - Construction of dams and reservoirs
 - Dikes and levees
-

13. Carbon Footprint

Q49. What is a carbon footprint?

Answer: The total amount of greenhouse gases released into the atmosphere due to human activities.

Q50. What are the major sources of carbon footprint?

Answer:

- Burning fossil fuels
- Transportation
- Industries
- Deforestation
- Thermal power plants

Q51. What are the effects of a high carbon footprint?

Answer:

- Global warming
- Climate change
- Sea-level rise
- Biodiversity loss
- Air pollution

Q52. How can students reduce carbon footprint?

Answer:

- Plant trees
- Save electricity
- Recycle materials
- Spread environmental awareness

EXTRA QUESTIONS

1. Explain the structure of the atmosphere and its layers.

The atmosphere is a thick blanket of gases surrounding the Earth. It is divided into five layers based on temperature changes.

1. Troposphere

- Lowest layer (8–18 km).
- Contains about 75% of atmospheric gases.
- All weather phenomena occur here.
- Supports life on Earth.

2. Stratosphere

- Extends up to about 50 km.
- Contains the ozone layer.
- Protects Earth from harmful UV rays.
- Suitable for aircraft flying.

3. Mesosphere

- Extends up to about 80 km.
- Meteorites burn up in this layer.

- Protects Earth from meteoroids.

4. Thermosphere (Ionosphere)

- Extends up to about 450 km.
- Helps in radio communication and satellite transmission.
- Hottest atmospheric layer.

5. Exosphere

- Outermost layer.
- Merges gradually with outer space.
- Satellites revolve in this region.

Conclusion: Each layer performs important functions that help sustain life and protect the Earth.

2. Describe the composition of the atmosphere.

The atmosphere is made up of gases, water vapour, and dust particles.

Major Gases

Nitrogen (78%)

- Most abundant gas.
- Essential for plant growth.
- Prevents rapid combustion.

Oxygen (21%)

- Necessary for respiration.
- Supports burning and combustion.

Argon (0.93%)

- Inactive gas.
- Used in electric bulbs and welding.

Carbon Dioxide (0.04%)

- Used in photosynthesis.

- Helps maintain Earth's temperature through the greenhouse effect.

Other Components

Water Vapour

- Forms clouds and rainfall.
- Controls humidity.

Dust Particles

- Help in cloud formation and rainfall.

Ozone

- Absorbs harmful ultraviolet rays.

Conclusion: All components of the atmosphere work together to support life and regulate climate.

3. Explain the elements of weather and climate.

Weather and climate are determined by five major elements.

1. Temperature

- Degree of hotness or coldness of air.
- Measured by a thermometer.

2. Atmospheric Pressure

- Force exerted by air on Earth's surface.
- Measured by a barometer.

3. Wind

- Movement of air from high pressure to low pressure areas.
- Speed measured by an anemometer.
- Direction measured by a wind vane.

4. Humidity

- Amount of water vapour present in air.

- Measured by a hygrometer.

5. Precipitation

- Rain, snow, hail, etc.
- Measured by a rain gauge.

Conclusion: These elements together determine the weather and climate of a region.

4. Describe the four seasons of India.

India experiences four major seasons.

1. Hot Weather Season (March–May)

- High temperatures.
- Low pressure develops over northern India.
- Hot winds called Loo blow in northern plains.
- Mango showers occur in Kerala and Karnataka.

2. Cold Weather Season (November–February)

- Cool and dry weather.
- Northeast trade winds prevail.
- Western Disturbances bring winter rainfall.

3. Advancing Monsoon Season (June–September)

- Southwest monsoon winds bring rainfall.
- Divided into Arabian Sea Branch and Bay of Bengal Branch.
- Main rainy season of India.

4. Retreating Monsoon Season (October–November)

- Monsoon winds withdraw.
- Weather becomes hot and humid (October Heat).
- Cyclones may affect eastern coastal regions.

Conclusion: These seasons influence agriculture, water resources, and daily life in India.

5. Explain the mechanism of the Southwest Monsoon.

The Southwest Monsoon is the main source of rainfall in India.

Mechanism

1. During summer, northern India becomes very hot.
2. A low-pressure area develops over northwestern India.
3. Southeast trade winds from the Southern Hemisphere cross the Equator.
4. Due to Earth's rotation, they become southwest winds.
5. These moisture-laden winds bring heavy rainfall to India.

Two Branches

Arabian Sea Branch

- Reaches Kerala first.
- Gives heavy rainfall to Western Ghats and western coast.

Bay of Bengal Branch

- Moves towards northeast India.
- Causes heavy rainfall in Assam, Meghalaya, and Ganga plains.

Importance

- Supports agriculture.
- Recharges rivers and reservoirs.
- Provides water for daily needs.

Conclusion: The Southwest Monsoon is known as the lifeline of India.

6. Discuss the causes and effects of climate change.

Causes of Climate Change

1. Burning fossil fuels.
2. Deforestation.
3. Industrial emissions.
4. Agricultural activities.
5. Excessive waste generation.

Effects of Climate Change

1. Rising global temperatures.
2. Melting glaciers and ice caps.
3. Rise in sea level.
4. Increase in floods, droughts, and storms.
5. Loss of biodiversity and wildlife habitats.

Human Impacts

- Food and water shortages.
- Health problems.
- Economic losses.

Conclusion: Climate change is a major global challenge that requires urgent action.

7. Explain the causes, effects and mitigation of floods.

Causes of Floods

1. Heavy rainfall.
2. Overflowing rivers.
3. Cyclones and storms.
4. Dam failures.
5. Tsunamis and glacier melting.

Effects of Floods

1. Loss of life and property.
2. Damage to crops.
3. Spread of diseases.
4. Soil erosion.
5. Displacement of people.

Mitigation Measures

1. Flood forecasting and warning systems.
2. Construction of dams and reservoirs.
3. Embankments, dikes, and levees.
4. Afforestation.
5. Proper land-use planning.

Conclusion: Effective planning and management can reduce flood damage significantly.

8. What is carbon footprint? Explain its sources and reduction methods.

Meaning

A carbon footprint is the total amount of greenhouse gases released into the atmosphere due to human activities.

Sources

1. Burning coal, oil, and natural gas.
2. Transportation.
3. Industries and factories.
4. Thermal power plants.
5. Deforestation.

Effects

1. Global warming.
2. Climate change.
3. Sea-level rise.
4. Air pollution.
5. Loss of biodiversity.

Reduction Methods

1. Use public transport.
2. Save electricity.
3. Use renewable energy.
4. Plant more trees.
5. Reduce, reuse, and recycle.

Conclusion: Reducing carbon footprint is essential for protecting the environment and future generations.